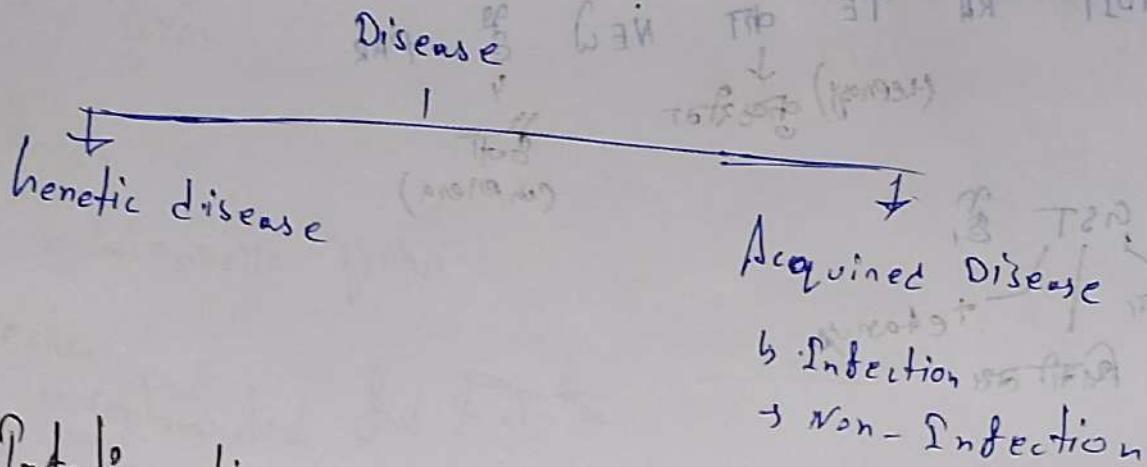


Human Disease



• Infection disease

- Virus
- Bacteria
- Protozoal
- Fungal

Bacterial disease

✓ ~~Typhoid~~ Typhoid

✓ Pneumonia

✓ Syphilis

✓ Plague

✓ Tuberculosis

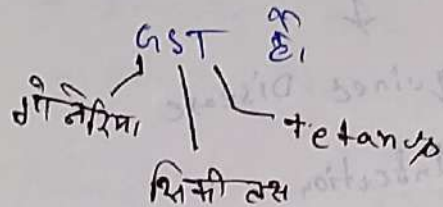
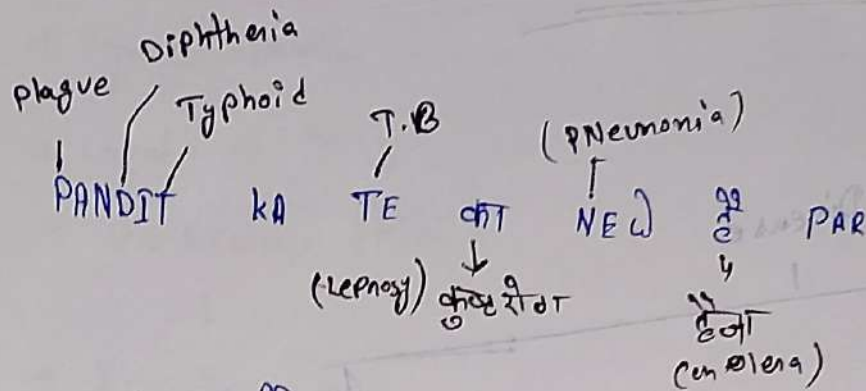
✓ Cholera

✓ Tetanus

✓ Leprosy

✓ Diphtheria

Learning trick of Bacterial diseases



बिमारी बनाने के दो तरह से फैलता है

• Pathogen (रोग जनक)

• Virus, Bacteria, etc.

• Vector

• Dog, cat, etc

• Pathogen -

• जो Virus Bacteria होता है

Vector -

• जो Carrier होता है उस Virus को
carry करता है

Typhoid

→ Enteric Fever

• Pathogen

↳ *Salmonella Typhi*

• Vector

↳ contaminated food & water

TEST :: WIDAL

Cholera

→ उदर - दस्त

• Pathogen

Vibrio cholerae

• Vector

↳ contaminated food & water

Pneumonia

• Pathogen

→ *Streptococcus / Diplococcus*

• Vector

→ Droplet

Plague

Red / Black spots on the body

• Pathogen

• *Yersinia pestis*

• Vector

• Rat

Tetanus

• Pathogen

→ *Clostridium Tetani*

• Vector

Soil / Dust / Open Wound

• It attacks CNS (Central Nervous System)

• Jaw lock

• *अस्थिज्वर* बيمारी

Vaccine

↳ DPT (Diphtheria, Pertussis, Tetanus)

Syphilis

• Pathogen

→ *Treponema Pallidum*

• Vector

↳ urinary track

↳ STD.

→ लसूना में है

• Initial symptoms, Lips cracks.

• Same Disease (Keratitis)

Tuberculosis

• Pathogen

→ *Mycobacterium Tuberculosis*

Vector → Air, water, food, soil.

Symptoms

→ Blood in cough

Vaccine

B.C.G. *Bacillus Calmette - Guérin*

Diphtheria

• Pathogen

→ Corynebacteria

• Vector

→ Air

Vaccine

→ DPT

Viral Disease

- Common cold
- pathogen
 ' Rhinovirus
- vector
 Air

DENHUE

- Pathogen
 Arbo family virus
- vector
 Female Aedes aegypti

- High fever
- Joints Pain
- Platelets

Smallpox

- Pathogen
 Variola virus
- vector

Vaccine

- Edward Jenner discover from cow serum
- Blood clotting factor

HIV / AIDS

• Pathogen.

↳ HIV (Human Immuno deficiency virus)

↳ Acquired Immuno deficiency Syndrome

• Vector

↳ mother to child

↳ STD

↳ Blood / Needle

WBC → Attack

HIV is virus

AIDS is disease

AIDS day - 1 Dec

Test

• ELISA : Enzyme linked Immuno Sorbant Assay

Drug (दवा) : AZT (AZidothymidine)

Symptoms

• हीट & कंठे होना

• भूख कम लगती

• वजन भी कम

Rabies

• Pathogen

- Rabies family

Vector

- Dog, cat, monkey

Hydrophobia (पानी से डर)

- mentally retired
- Skin

Mumps (गाल बुझा)

Pathogen

- Paramyxovirus

vector

air / water / soil

It attacks salivary gland

गले या सूजन आ जाती है

Tired Bones

Digestive system affected

Ebola

Pathogen

→ Ebola virus

(Ebola is a name of River)

vector

→ Bat.

RBC start decreasing

Diarhoea

Anemia (रक्त की कमी)

weakness in hands & legs

Nipah

Pathogen

Nipah virus

(zoonotic virus) → (Animal to human)

Brain fever

Brain dead

coma

Pathogen

Swine flu

↳ H₁N₁ / H₃N₂

Vector is Swine / pig

जो भी भोजन

उली

Bird flu

Pathogen

↳ H₅N₁

Vector

Ducks, hens

~~Ris effect~~ Respiratory system affected

Immunity ↓

Hepatitis - A

Yellow Jaundice

• Hepatitis - A Virus

- Liver cell damage
- Bilirubin (रक्त) में बढ़ा देता है
- पीलिया

Hepatitis - B

White Jaundice

• Hepatitis B Virus

→ रक्त का पानी

- High level Anemia
- AIN
- Blood Infection

SARS

- Severe Acute Respiratory System
- 1st case within (2002)

Symptoms

- Cold
- fever
- Respiratory System Affected
- Communicable

Measles

Pathogen

- Paramyxovirus

Vector

Water, food, Air.

Symptoms

- आँखों में लाल
- Blood in excretion

German measles

- Rubella virus.
- RBC घटा होता है
- Respiratory system भी प्रभावित होता है
- ~~Diff~~ Difficulty in breathing

Influenza

- Influenza virus
- Air

Symptoms

- Cold
- Fever
- Cough
- Headache

Protozoal Disease

Malaria

- Plasmodium. virus

Vector

Female Anopheles

Symptoms

- तेज बुखार
- RBC कम हो जाता है
- Liver infection

दवा

~~Quinine~~ Quinine

Quinine is made with Bark of cinchona

25 April मलेरिया दिवस

Sleeping sickness / African Sleeping Sickness

• Trypanosoma / Trypanosoma virus

• Tse - Tse Fly

• जीवा वृद्ध हो जाता है

• Nervous System पे Attack करता है

Elephantiasis

- Wuchereria bancrofti (virus)
- Vector. Female Culex

Symptoms

एक पैर पर सूजन आ जाता है।

KALA-AZAR

- Leishmania donovani ~~virus~~ Protozoa
- Sand fly.
- तेज बुखार
- Mentally ill

Diarrhea

- Entamoeba histolytica, virus
- Liquid.
- Small intestine

उल्टी, दस्त

Pyorrhoea

- Entamoeba gingivitis
- मसूड़ों से रक्त आना शुरू हो जाता है
- Smell (बुराई)
- मसूड़ी पर सूजन

Fungal diseases

Ringworm (T.T.).

- Trichophyton
- Microsporum
- Epidermophyton

} fungus by which it is caused

Trichophyton

Microsporum

Scabies

- Aspergillus fumigatus / Ascaris fungus.
- Body Red + Itching

Baldness

- Tinea capitis (Dandruff)

• Athlete's foot

• Tinea Pedis.

It is a fungal infection of the skin.

Asthma

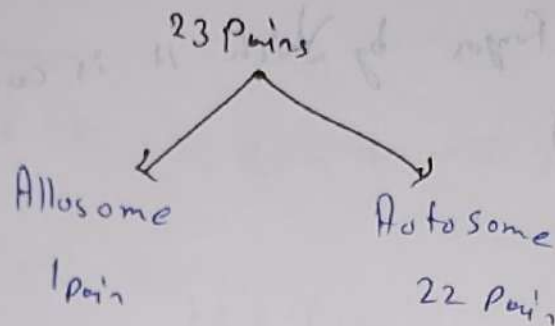
→ Aspergillus fumigatus fungus

• Trachea

Inflamed or + HE

Genetic diseases

- Genes are bits of DNA



Colour blindness

↳ R.G.B. में colour नहीं दिखता है
उसे ~~ही~~ दिखता है

- In male $\rightarrow$ 8% chance of it
- In female $\rightarrow$ 0.4% "

X linked chromosomal disease.

Hemophilia

- रक्त का थक्का नहीं लगता
- X-linked chromosomal disease
- Royal disease (also known as)
- Because $\rightarrow$ British $\rightarrow$ Queen Victoria
- ↳ Christmas disease (25 Dec)
- It was found on

(Christmas factor)

Factor VIII



Hemophilia



Absent

Factor IX



Hemophilia B



Absent

Queen Victoria have Hemophilia B

Sickle cell Anemia:

- रक्त का थक्का नहीं बनता
- Iron की कमी
- Both ~~Male~~ Male & Female
- Oxygen की कमी

Thalassemia

- रक्त की कमी
- RBC dies before 120 days
- Hemoglobin कम होता है

Klinefelter Syndrome

→ Allosomes

→ $22 + X, Y$

$44 + X, X, Y \rightarrow 47$ chromosomes

• cluster of disease

Down Syndrome

• male + female

• Autosome

• 21th pair

Trisomy

Trisomy in chromosome.

↳ $A + B + C$

• मंडा वृद्ध

Turner Syndrome

• Disorder in female Allosomes

$22 + X (-) = 45$

↳ ovaries

↓

• Sterility. (वैश्मयन)

Palato Syndrome

13th pair chromosomes disorder

trisomy

→ 3rd pair 13th pair

→ Digestion

→ Both (Male + Female)

Edward Syndrome

Both (male + female)

18th pair chromosome
trisomy

→ finger overlapping

→ weight loss

Non-Infectious Disease

Cancer

cell division

→ uncontrolled division

→ new cell (collect)
↳ (Tumour)

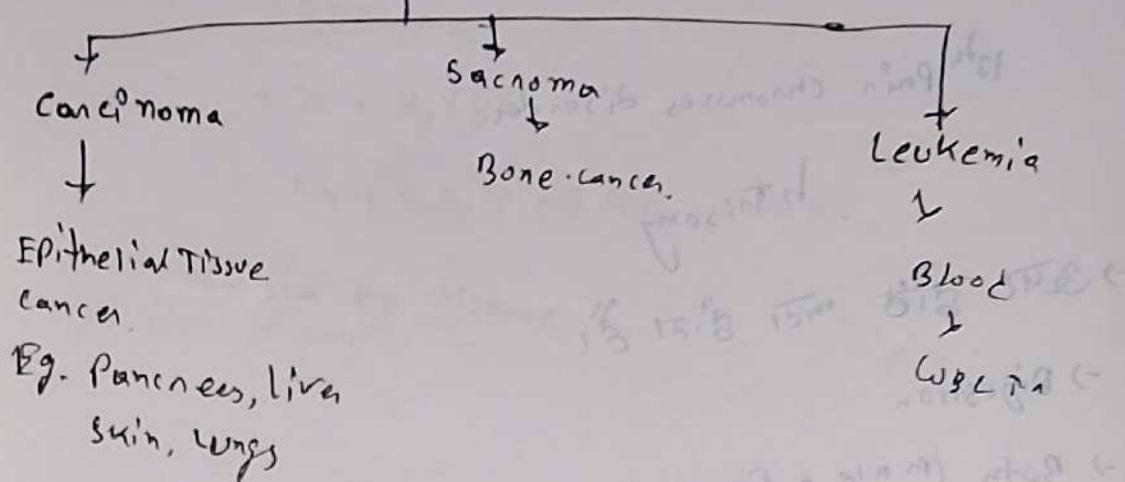
Tumour

Benign

Malignant

metastasis

Types of Cancer



Reason

- Radiation
- UV Rays
- Tobacco
- Nails
- Coal mine
- Cement factory
- Cotton factory
- ~~Paint~~ Paint factory

• Biopsy

↳ Piece
↳ Tissue
Tested in Lab.

• MRI → Magnetic Resonance Imaging

↳ Exact Part

• CT-Scan → Computer Tomography
↳ 3D Image

Treatment

• Surgery

- ↳ Tumor
- ↳ remove

• chemotherapy

- ↳ Drugs
- ↳ cell division

• Radiation

- ↳ UV Rays

Hair - Hair

• Nutrition

- ↳ CO_2

• Back bone → Bend

• Joints Pain

- chocolate & Cd^{2+} Cadmium

Dermatitis